QP CODE:
25900019

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION MARCH 2025
SECOND SEMESTER

Discipline Specific Core Course (DSC) - MG2DSCBOT100, MG2DSCBBT100- PLANT RESOURCES AND VENTURES IN BOTANY
(2024 ADMISSION ONWARDS)

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)
Students should attempt at least one question from each course outcome to enhance their overall
outcome attainability

Duration : 1.5 Hours Max. Marks : 50

Part A -
Short Answer- Answer any 10 questions - Each question carries 1 marks

1	What are biocontrol agents ?	[K] / [CO1]
	Biocontrol agents are living organisms, like natural enemies of pests and pathogens, used to manage unwanted organisms in a sustainable way. These agents can include predators, parasites, pathogens (bacteria, fungi, viruses), and other natural enemies that help control the population of pests and diseases.	
2	Name the plant group which Pacific Yew belongs to.	[K] / [CO1]
	The Pacific Yew, scientifically known as *Taxus brevifolia*, belongs to the plant group Taxaceae, also known as the yew family. **Or** Gymnosperms Or Drug yielding plant	
3	Name one example of a plant-based nutraceutical.	[K] / [CO1]
	popular plant-based nutraceuticals include turmeric (for its anti-inflammatory properties), Boswellia and glucosamine (for joint health), eucalyptus and peppermint (for respiratory health), and natural fibers and probiotics (for digestive health) (or any other popular ones)	
4	Which plant part is used for manufacturing Coir?	[K] / [CO1]
	Coir is manufactured from the mesocarp of the coconut fruit, which is the fibrous outer layer also known as the husk.(**marks can be given to – husk or sclerenchyma fibers**)	
5	Who is a Nursery manager?	[U] / [CO5]
	A Nursery Manager is someone who oversees the operations of a plant nursery, focusing on the care, growth, and development of plants. They are responsible for tasks like planting, pruning, cultivating, and harvesting seedlings, as well as managing staff and resources	
6	Name any two crop plants researched by CTCRI.	[U] / [CO2]

		Tapioca, Yams, Arum, Potato, sweet potato etc (any two)	
7	Expand FCI		[U] / [CO2]
		Food Corporation of India (out of syllabus – full credit if attended and written anything related to food)	
8	What is CRISPR-Cas9?		[U] / [CO2]
		CRISPR-Cas9 is a gene-editing technology that allows scientists to precisely change DNA sequences. (**out of syllabus – full credit if attended and written anything related to molecular biology**)	
9	Give any two examples for gardens		[U] / [CO3]
		TBGRI, Government Botanical garden Ooty, Lalbagh, Brindhavan, Tajmahal garden, Moghul garden, Delhi (or any two famous gardes)	
10	Expand SSC.		[K] / [CO5]
		Staff Selection Commission	
11	Give an example for financial assistance schemes for entrepreneurs		[U] / [CO4]
		MSME, IRDP, KVIB, Horticulture mission, Coconut Development Board (any other organization offering schemes	
12	Name a biotechnology firm functioning in Kerala		[U] / [CO5]
		ABTECH, Chingavanam, Kottayam AVT Biotechnology (L.J. International Ltd.): ZAARA Biotech (any other famous biotech based firms)	
		(1x10 = 10)	

Part B
Short Essay - Answer any 6 questions - Each question carries 5 marks

13	How do plants improve soil fertility?		[U] / [CO1]
		Plants improve soil fertility through several mechanisms **Adding Organic Matter**: - Plant parts decompose, adding organic matter to the soil. This organic matter improves soil structure, making it more porous and allowing for better water infiltration and aeration. **Nitrogen Fixation**: Legumes, like beans and peas, have a symbiotic relationship with bacteria that can convert atmospheric nitrogen into a form plants can use. **Improving Soil Structure**: Plants' roots help bind soil particles together, creating aggregates that improve soil structure and enhance its ability to hold water and nutrients. **Nutrient Uptake and Cycling**: Plants take up nutrients from the soil and then release them when they decompose. This nutrient cycling helps maintain soil fertility.	

		Crop Rotation: Rotating different crops can help prevent the depletion of specific nutrients in the soil, as different plants have different nutrient needs. For example, some crops can increase potassium levels in the soil, which can then be used by other crops. **Cover Cropping**: Growing cover crops, like legumes, grasses, and clovers, during the off-season can protect the soil from erosion, add organic matter, and even fix nitrogen from the atmosphere. **Beneficial Microorganisms**: Plants can also promote the growth of beneficial soil microorganisms that contribute to soil health and fertility. (**5 points = full credit**)	
14	Write the herbal and cosmetic uses of Bringaraj.		[K] / [CO1]
		Bhringraj (Elipta alba, Eclipta prostrata) is an herb with numerous herbal and cosmetic uses Herbal Uses: Hair Growth and Hair Fall: popular Ayurvedic remedy for hair growth and to reduce hair fall. It nourishes the scalp, improves blood circulation, and strengthens hair follicles. **Prevents Premature Graying**: can help prevent or delay the onset of gray hair. **Scalp Health**: can help reduce dandruff, scalp irritation, and fungal infections. **Digestive Health**: In Ayurveda, Bhringraj is believed to stimulate digestion and aid in nutrient absorption. **Liver Health**: traditionally used to support liver function. **Eye Health**: sometimes used as an eye tonic to improve eyesight. **Cosmetic Uses:** **Skin Conditions**: anti-inflammatory and antibacterial properties can be helpful for skin conditions like acne, eczema, and psoriasis. **Acne:** can help reduce acne by fighting bacteria and soothing inflammation. **Dry Skin**: can help moisturize and soothe dry skin. A**geing**: its antioxidant properties may help slow down the aging process by reducing wrinkles and fine lines. **Skin Radiance**: can help improve skin tone and radiance by promoting a healthy glow. (**specific herbal use 2 marks, specific cosmetic use -3 marks**)	
15	Give binomial, families and plant parts of any 3 plants used as vegan cosmetics. Explain why vegan cosmetics are considered to be significant.		[U] / [CO1]
		Plants used in Vegan Cosmetics Many plants are used in vegan cosmetics due to their moisturizing, soothing, and antioxidant properties. Some popular examples include aloe vera, coconut oil, green tea, chamomile, lavender, and rosemary. Other plants used include neem, hibiscus, and various oils like jojoba, almond, and argan oil. **Aloe vera** Binomial - Aloe barbadensis miller. Family - Asphodelaceae (Liliaceae) Plant par used – Gel from succulent leaf **Coconut oil** Binomial – *Cocos nucifera*.	

		Family - Arecaceae Plant par used – Kernel oil **Green Tea** Binomial - Camellia sinensis (Thea sinensis, or C. thea) Family - Theaceae,Asphodelaceae (Liliaceae) Plant par used – cured dried tender leaves **Chamomile** Binomial - Matricaria recutita (syn. Matricaria chamomilla, Chamomilla recutita (L.) Family - Asteraceae Plant par used – Essential oil distilled from flowers **Lavender** Binomial - Lavandula spica Family - Lamiaceae Plant par used – Essential oil distilled from flowers **Rosemary** Binomial - Rosmarinus officinalis L. (syn. Salvia rosmarinus Spenn.), Family - Lamiaceae Plant par used – Leaves **Vegan cosmetics are significant** because they offer an ethical choice for consumers concerned about animal welfare and cruelty-free practices in the beauty industry. They also prioritize natural and plant-based ingredients, which can be gentler on the skin and reduce the risk of allergic reactions. Furthermore, vegan cosmetics often have a lower environmental impact compared to their non-vegan counterparts, using sustainably sourced ingredients and reducing the need for animal-derived products. (**botanical name, family and part used – 3 marks, Definition and significance – 2 marks**)	
16	How does market analysis helps an entrepreneur?		[U] / [CO4]
		Market analysis benefits entrepreneurs: - **1. Understanding the Target Audience:** Identifying Needs and Preferences: Market research helps to understand their target audience's needs, preferences, and buying habits. Effective Marketing: - Understanding the target audience enables entrepreneurs to develop effective marketing campaigns and messaging. **2. Identifying Market Trends and Opportunities:** Spotting New Trends: - Market analysis helps identify emerging trends, changes in customer behavior, and technological advancements.	

		Capitalizing on Opportunities: This information allows entrepreneurs to adapt to market changes and identify new opportunities for growth and innovation. Reducing Risks: By understanding the market, entrepreneurs can minimize risks associated with launching new products or services. **3. Analyzing the Competitive Landscape:** Identifying Competitors: Market analysis helps entrepreneurs identify their competitors and understand their strengths and weaknesses. Differentiating Their Business: This knowledge allows entrepreneurs to differentiate their business and create a unique value proposition. Developing Competitive Strategies: - Understanding the competitive landscape helps entrepreneurs develop strategies to gain a competitive edge. **4. Reducing Risks and Costs:** Validating Business Ideas: Market research helps entrepreneurs validate their business ideas and assess their potential for success. Identifying Potential Barriers: It helps identify potential barriers to entry, such as regulations or competitive pressures. Minimizing Costs : - By reducing the risk of launching a business that doesn't meet market needs, market analysis helps entrepreneurs minimize costs. **5. Making Informed Decisions:** Strategic Planning: Market analysis provides the data and insights needed to make informed decisions about business strategy, pricing, and product development. Resource Allocation: - It helps entrepreneurs allocate resources effectively by focusing on the most promising opportunities and target markets. Improving Business Performance: - By making informed decisions based on market analysis, entrepreneurs can improve business performance and achieve better outcomes . (**any three with brief explanation – full credit**)	
17	What is layering? Name any two types of layering.		[K] / [CO3]
		Layering is a plant propagation method where stems or branches are encouraged to root while still attached to the parent plant. There are several types of layering, including simple, compound (serpentine), mound (stool), tip, and air layering, each with a slightly different approach. Types of Layering: Simple Layering: A stem is bent to the ground and buried in soil, encouraging root development. Compound (Serpentine) Layering: Multiple points along a stem are buried to create several new plants from a single stem. Mound (Stool) Layering: The base of a plant is cut back, and soil is piled over the new shoots to promote rooting. Tip Layering: Flexible stems are bent and the tips are buried in soil. Air Layering: Roots are formed on a stem above ground, usually by wrapping a section of the stem in a moist medium like sphagnum moss and plastic.	

		(**layering- definition – 1 marks, Types with method in brief – 2 marks each**)	
18	What are the key achievements of the National Rice Research Institute?		[U] / [CO2]
		Key Achievements of the Indian Rice Research Institute (IIRR): **1. <u>Development and Release of Rice Varieties</u>**: Rice varieties with enhanced nutritional content and resistance to various stresses. High-Protein Rice Variety (CR Dhan 310): Improved Samba Mahsuri: This is a popular, low-glycemic index rice variety DRRH 3: the first high-yielding rice hybrid with medium slender (MS) grain **<u>System of Rice Intensification (SRI):</u>** IIRR is a success story in promoting SRI, a water-saving and high-yielding rice cultivation technique. **<u>Site-Specific Nutrient Management</u>**: IIRR has contributed to the development and implementation of site-specific nutrient management practices in rice farming. <u>Other Achievements</u>: IIRR has also made advancements in areas like trap crop management for yellow stem borer, drum seeder technology, and hybrid purity testing. **Collaboration with other institutions:** IIRR collaborates with other research institutions, agricultural universities, and the Indian Council of Agricultural Research (ICAR) to promote rice research and development. **Focus on Sustainable Rice Farming**- IIRR emphasizes sustainable rice farming practices, including water management, nutrient management, and pest management, according to the Indian Council of Agricultural Research. (**min. 3 achievements with example = full credit**))	
19	Explain the significance of entrepreneurship in the context of plant science.		[K] / [CO4]
		Plant-based entrepreneurship is significant due to its potential to address growing health concerns, environmental issues, and consumer demand for sustainable and cruelty-free alternatives. **<u>1. Addressing Health Concerns:</u>** Reduced Health Risks: Plant-based diets are often associated with reduced risks of heart disease, type 2 diabetes, and certain cancers. Improved Gut Health: Plant-based foods are rich in fiber, which supports a healthy gut microbiome, aiding in digestion and nutrient absorption. **<u>2. Environmental Benefits:</u>** Reduced Carbon Footprint: Plant-based food production requires significantly less water, land, and energy compared to animal agriculture, contributing to a lower carbon footprint. Reduced Water Pollution: Lower Greenhouse Gas Emissions: Plant-based meat production emits up to 94% less greenhouse gas emissions than animal-based meat production, says The Good Food Institute. Biodiversity Protection: By reducing the need for animal agriculture, plant-based	

		entrepreneurship can help protect ecosystems and biodiversity. **3. Economic Opportunities**: Growing Market: The global plant-based food market is experiencing significant growth, Investment and Innovation: Venture capital investments in plant-based food companies are increasing, driving innovation and new product development. Job Creation: The plant-based food industry is creating new jobs in food processing, manufacturing, and distribution. **4. Ethical Considerations:** Animal Welfare: Plant-based entrepreneurship aligns with growing concerns about animal welfare and ethical consumption. Sustainability: It promotes sustainable food systems that minimize environmental impact and resource depletion. **5. Cultural Impact:** Shifting Food Culture: The rise of plant-based food is influencing food culture and consumer preferences, with more people adopting plant-based diets.	
20	Describe the different roles of botanists.		[K] / [CO5]
		1. Research and Academic Roles: Plant Identification and Classification: Plant Physiology and Genetics: study plant processes as well as the genetic basis of plant traits. Ecology and Environmental Science - contributing to understanding ecosystems and addressing environmental issues. Plant Pathology: They study plant diseases, developing strategies to protect crops and maintain plant health. **2. Conservation and Restoration:** Endangered Species Protection: work to conserve threatened plant species, often through habitat preservation, breeding programs, and reintroduction efforts. Ecosystem Restoration: help restore degraded ecosystems, using their knowledge of plant ecology and plant communities to re-establish native vegetation. Environmental Consultancy: provide expertise in environmental impact assessments, land management, and sustainable development planning. **3. Applied Sciences:** Agriculture and Crop Science: develop improved crop varieties, optimize farming practices, and address issues like crop diseases and pest control. Pharmaceuticals and Biotechnology: - research plant-based medicines, explore new therapeutic applications, and contribute to the development of plant-based biotechnologies. Environmental Remediation: develop strategies for cleaning up contaminated sites, using plants to absorb pollutants or stabilize soil. **4. Education and Outreach:**	

		Teaching and Mentoring: teach botany courses at universities and colleges, inspiring the next generation of plant scientists. Public Education and Outreach: share their knowledge with the public through museum exhibits, botanical gardens, and public lectures, promoting plant appreciation and conservation. 5**. Other Roles:** Naturalists: often work as naturalists, studying plant life in its natural habitat and educating the public. Land Managers: They manage land resources, ensuring sustainable use of plants and ecosystems. Agronomists: apply botanical knowledge to optimize agricultural practices and improve crop yields. **(any three in details – full credit)**
		(5 × 6 = 30)

Part C **Essay Answer any 1 questions - Each question carries 10 marks**		
21	Explain different types of value added products from fruits and their uses.	[U] / [CO1]
		Value-added fruit products are raw fruit transformed into a more marketable form, often with extended shelf life. Common examples include jams, jellies, juices, pickles, dried fruit, chips, and halwa. Specific products can be customized to suit the fruit's characteristics, 1. Jams and Jellies: Using sugar and pectin to preserve the fruit in a spreadable form. 2. Juices and Nectars: Extracting juice and potentially adding pulp or flavors for different textures and tastes. 3. Dried Fruit: Removing moisture to extend shelf life, such as dried mango slices or banana chips. 4. Pickles: Preserving fruits in a brine solution with spices, like mango pickles. 5. Canned Fruits: Preserving fruits in their own juice or syrup to maintain their texture and flavor. 6. Frozen Fruits: Quickly freezing fruits for later use, often as a component of other products. Specific Product Examples: 1. Mango: Mango pulp, concentrate, jams, jellies, pickles, juices, and dried slices. 2. Banana: Banana chips, banana bar, banana puree, and banana flour for biscuits. 3. Citrus Fruits: Juices, concentrates, jams, marmalades, and pickles. 4. Grapes: Wine, juice, and dried raisins. 5. Pineapple: Juice, wine, vinegar, and textile fibers from the leaves. 6. Jackfruit: Chips, papads, pickles, ice cream, jelly, and sweets. 7. Avocado: Oil, flour, and pulp for food ingredients.

		8. Tomatoes: Puree, paste, sauces, and dried slices. 9. Other: Fruit halwa, fruit candy, and baby food. (**min 5 products with method of value addition and example = 10 marks**)	
22		Write an essay on organic farming	[U] / [CO3]
		Organic farming is a system of agriculture that uses natural processes and resources to produce food, prioritizing healthy soil, beneficial insects, and a thriving ecosystem. It avoids synthetic pesticides, fertilizers, and genetically modified organisms (GMOs). Instead, it relies on organic materials like crop residues, animal manures, and biological pest control methods. **<u>Key Practices of organic farming:</u>** : Soil Health: - building and maintaining healthy soil through practices like crop rotation, cover cropping, composting, and minimal tillage. Pest Control: Instead of synthetic pesticides, organic farmers use natural methods like beneficial insects, crop rotations, and resistant varieties to control pests. Fertilizers: - use organic materials like compost, animal manure, and green manure to provide nutrients to crops, rather than synthetic fertilizers. Water Management: - incorporates water conservation practices to ensure efficient use and prevent runoff. Crop Rotation: - Rotating crops helps to improve soil health, control pests and diseases, and reduce the need for synthetic inputs. **<u>Benefits of Organic Farming</u>**: Healthier Food: - can produce food with higher nutrient content and fewer synthetic residues. Environmental Protection: - By avoiding synthetic chemicals, organic farming helps to protect soil, water, and air quality, and promotes biodiversity. Reduced Pollution: - Organic practices reduce the risk of pesticide and fertilizer runoff, which can pollute waterways and harm aquatic life. Improved Soil Health: - help to improve soil fertility, structure, and water-holding capacity. Biodiversity: - promotes biodiversity by creating habitats for beneficial insects and other wildlife. **<u>Challenges of Organic Farming:</u>** Lower Yields:- may result in lower yields compared to conventional farming, as it avoids the use of synthetic inputs that can boost crop production. Higher Costs: - can be more expensive due to the cost of certified organic products and the labor involved in implementing organic practices. Transition Period: - Farms transitioning to organic farming may experience a period of lower yields as they rebuild their soil and establish organic practices (**organic framing- practices adopted = 6 marks, advantages -2 marks, challenges 2**	

		marks)
		(10 × 1 = 10)